

Use the Lan stress test automation tool(LAN stress for iperf and dp_tcp.zip) to perform Lan card stress test

1.測試目標 (Test Objective)

-Windows

步驟:

步驟 1~5 Server 與 Client 都需執行

1)將需要測試的 Lan card port 手動設置為固定 ip

Server 端為 192.168.x.1

Client 端為 192.168.x.2

2)將 LAN stress for iperf and dp_tcp.zip 放置到 Windows 的環境裡

3)將 LAN stress for iperf and dp_tcp.zip 解壓縮

4)打開 LAN_stress_win 資料夾

5)執行 LAN-stress.bat

6)進行 iperf 測試

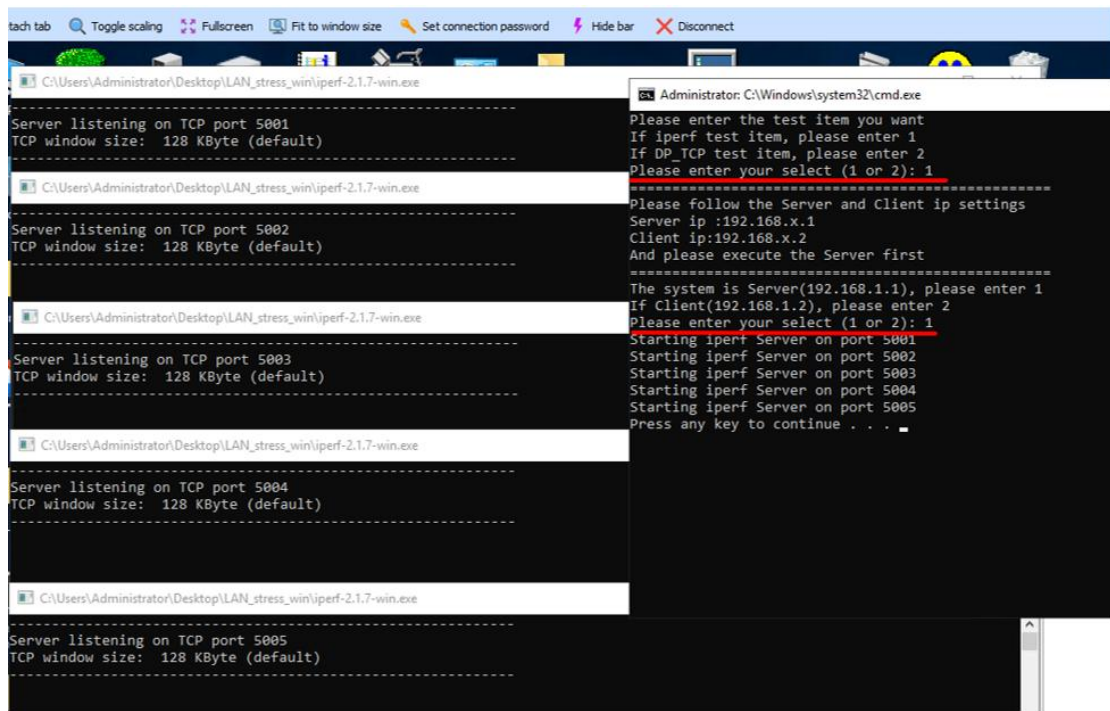
Server 端

A.根據下圖可得知選擇"1"即可進行 iperf 測試

B.根據下圖可得知選擇"1"為 Server 端

```
Administrator: C:\Windows\system32\cmd.exe
Please enter the test item you want
If iperf test item, please enter 1
If DP_TCP test item, please enter 2
Please enter your select (1 or 2): 1
=====
Please follow the Server and Client ip settings
Server ip :192.168.x.1
Client ip:192.168.x.2
And please execute the Server first
=====
The system is Server(192.168.1.1), please enter 1
If Client(192.168.1.2), please enter 2
Please enter your select (1 or 2): 1
Starting iperf Server on port 5001
Starting iperf Server on port 5002
Starting iperf Server on port 5003
Starting iperf Server on port 5004
Starting iperf Server on port 5005
```

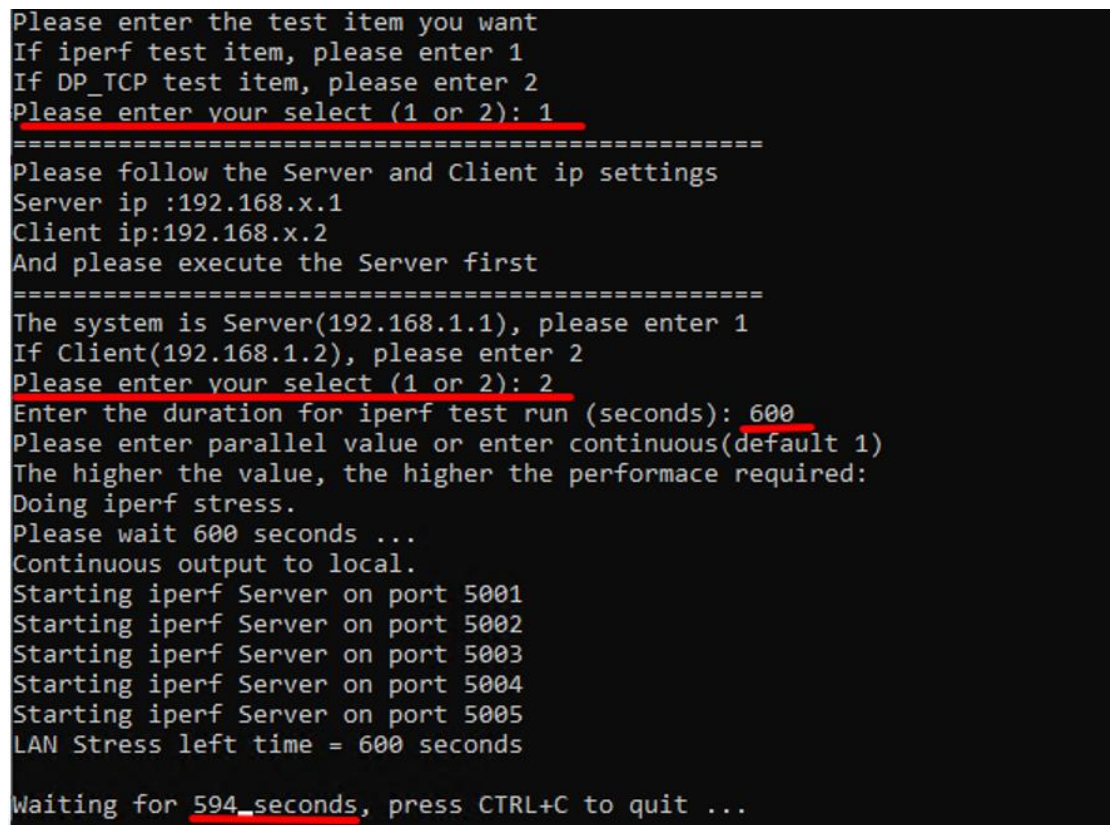
C.看到圖片二畫面即為完成 Server 端設定



Client 端

A.根據下圖可得知選擇"1"即可進行 iperf 測試

B.根據下圖可得知選擇"2"為 Client 端



C.輸入需測試時間(以秒為單位)

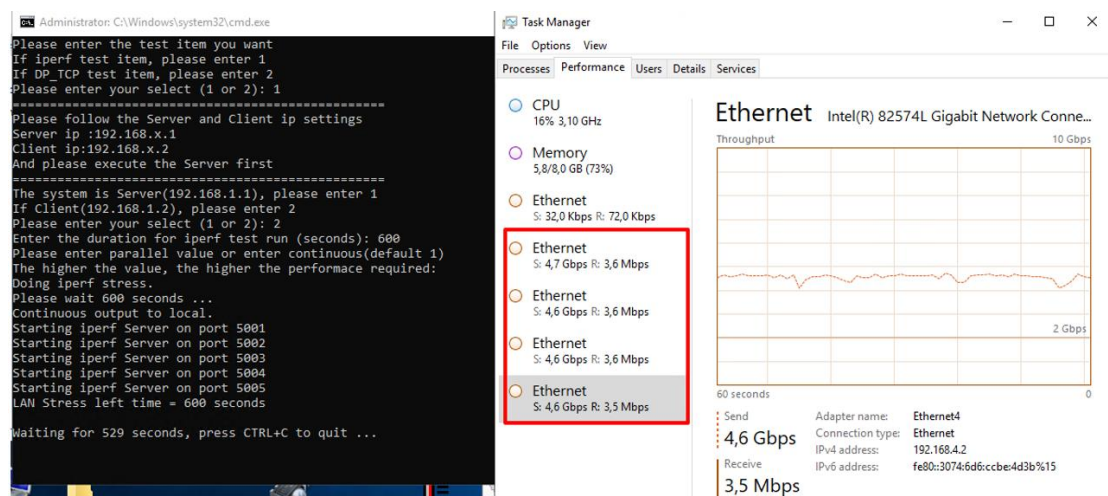
D.看到下圖畫面即為完成 Client 端設定(會出現倒數即為正在測試中)

```
Administrator: C:\Windows\system32\cmd.exe

Please enter the test item you want
If iperf test item, please enter 1
If DP_TCP test item, please enter 2
Please enter your select (1 or 2): 1
=====
Please follow the Server and Client ip settings
Server ip :192.168.x.1
Client ip:192.168.x.2
And please execute the Server first
=====
The system is Server(192.168.1.1), please enter 1
If Client(192.168.1.2), please enter 2
Please enter your select (1 or 2): 2
Enter the duration for iperf test run (seconds): 600
Please enter parallel value or enter continuous(default 1)
The higher the value, the higher the performance required:
Doing iperf stress.
Please wait 600 seconds ...
Continuous output to local.
Starting iperf Server on port 5001
Starting iperf Server on port 5002
Starting iperf Server on port 5003
Starting iperf Server on port 5004
Starting iperf Server on port 5005
LAN Stress left time = 600 seconds

Waiting for 529 seconds, press CTRL+C to quit ...
```

E.手動打開工作管理員來查看是否正在執行測試(可參考圖片五)



F.測試結束後 log 會放置在"iperf_log"裡(可參考圖片六)

LAN_stress_win > iperf_log		Search iperf_log	
Name	Date modified	Type	Size
192.168.1.1.txt	21.02.2025 18:23	Text Document	32 KB
192.168.2.1.txt	21.02.2025 18:23	Text Document	32 KB
192.168.3.1.txt	21.02.2025 18:23	Text Document	32 KB
192.168.4.1.txt	21.02.2025 18:23	Text Document	32 KB
192.168.5.1.txt	21.02.2025 18:13	Text Document	1 KB

7)進行 DP_TCP 測試

Server 端

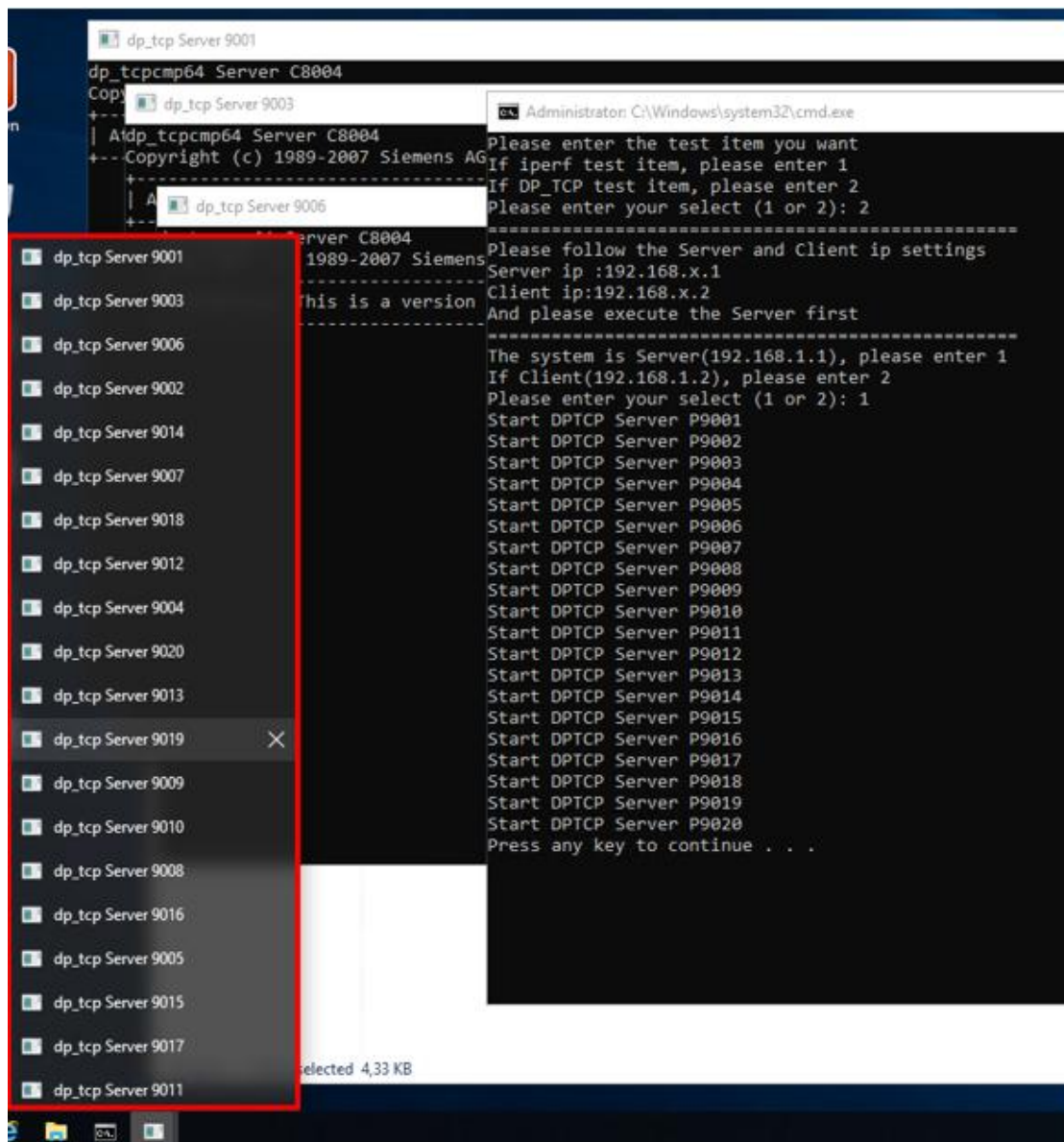
A.根據圖片七可得知選擇"2"即可進行 DP_TCP 測試

B.根據圖片七可得知選擇"1"為 Server 端

Administrator: C:\Windows\system32\cmd.exe

```
Please enter the test item you want
If iperf test item, please enter 1
If DP_TCP test item, please enter 2
Please enter your select (1 or 2): 2
=====
Please follow the Server and Client ip settings
Server ip :192.168.x.1
Client ip:192.168.x.2
And please execute the Server first
=====
The system is Server(192.168.1.1), please enter 1
If Client(192.168.1.2), please enter 2
Please enter your select (1 or 2): 1
Start DPTCP Server P9001
Start DPTCP Server P9002
Start DPTCP Server P9003
Start DPTCP Server P9004
Start DPTCP Server P9005
Start DPTCP Server P9006
Start DPTCP Server P9007
Start DPTCP Server P9008
Start DPTCP Server P9009
Start DPTCP Server P9010
Start DPTCP Server P9011
Start DPTCP Server P9012
Start DPTCP Server P9013
Start DPTCP Server P9014
Start DPTCP Server P9015
Start DPTCP Server P9016
Start DPTCP Server P9017
Start DPTCP Server P9018
Start DPTCP Server P9019
Start DPTCP Server P9020
```

C.看到圖片八畫面即為完成 Server 端設定



Client 端

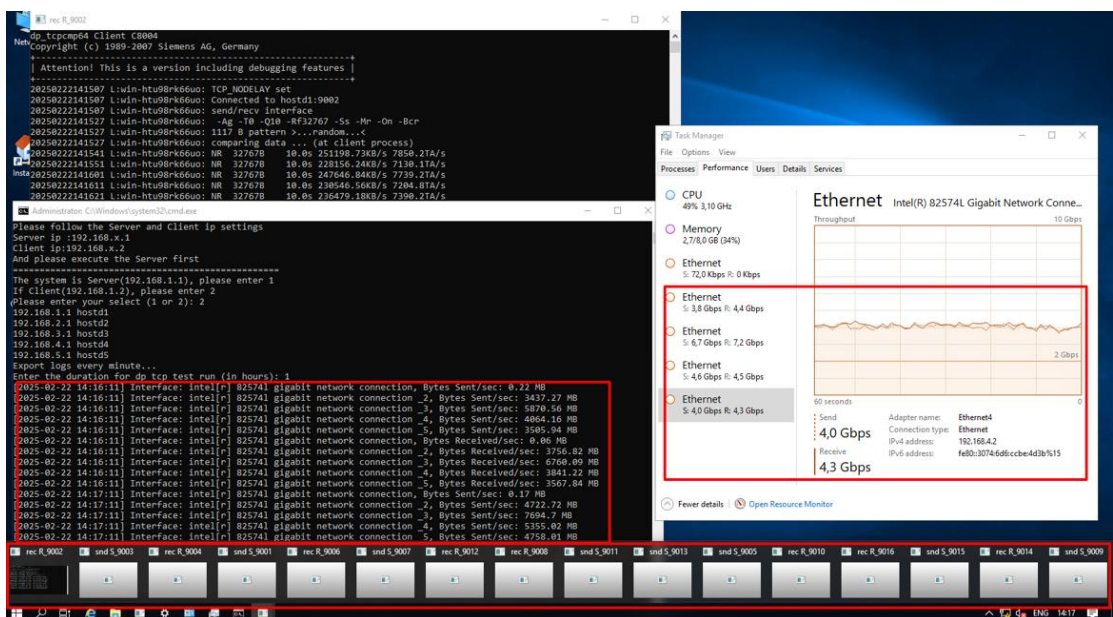
A.根據圖片九可得知選擇"2"即可進行 DP_TCP 測試

B.根據圖片九可得知選擇"2"為 Client 端

```
Administrator: C:\Windows\system32\cmd.exe

Please enter the test item you want
If iperf test item, please enter 1
If DP_TCP test item, please enter 2
Please enter your select (1 or 2): 2
=====
Please follow the Server and Client ip settings
Server ip :192.168.x.1
Client ip:192.168.x.2
And please execute the Server first
=====
The system is Server(192.168.1.1), please enter 1
If Client(192.168.1.2), please enter 2
Please enter your select (1 or 2): 2
192.168.1.1 hostd1
192.168.2.1 hostd2
192.168.3.1 hostd3
192.168.4.1 hostd4
192.168.5.1 hostd5
Export logs every minute...
Enter the duration for dp_tcp test run (in hours): 1
[2025-02-22 14:16:11] Interface: intel[r] 825741 gigabit network connection, Bytes Sent/sec: 0.22 MB
[2025-02-22 14:16:11] Interface: intel[r] 825741 gigabit network connection _2, Bytes Sent/sec: 3437.27 MB
[2025-02-22 14:16:11] Interface: intel[r] 825741 gigabit network connection _3, Bytes Sent/sec: 5870.56 MB
[2025-02-22 14:16:11] Interface: intel[r] 825741 gigabit network connection _4, Bytes Sent/sec: 4064.16 MB
[2025-02-22 14:16:11] Interface: intel[r] 825741 gigabit network connection _5, Bytes Sent/sec: 3505.94 MB
[2025-02-22 14:16:11] Interface: intel[r] 825741 gigabit network connection, Bytes Received/sec: 0.06 MB
[2025-02-22 14:16:11] Interface: intel[r] 825741 gigabit network connection _2, Bytes Received/sec: 3756.82 MB
[2025-02-22 14:16:11] Interface: intel[r] 825741 gigabit network connection _3, Bytes Received/sec: 6760.09 MB
[2025-02-22 14:16:11] Interface: intel[r] 825741 gigabit network connection _4, Bytes Received/sec: 3841.22 MB
[2025-02-22 14:16:11] Interface: intel[r] 825741 gigabit network connection _5, Bytes Received/sec: 3567.84 MB
```

- C.輸入需測試時間(以小時為單位)
- D.看到圖片十畫面即為完成 Client 端設定
- E.手動打開工作管理員來查看是否正在執行測試(可參考圖片十)



F.測試結束後 log 會放置在"LAN_stress_win"裡，檔名為 dp_tcp_win.log

-Linux

步驟:

步驟 1~5 Server 與 Client 都需執行

1)將需要測試的 Lan card port 手動設置為固定 ip

Server 端為 192.168.x.1

Client 端為 192.168.x.2

2)將 LAN stress for iperf and dp_tcp.zip 放置到 Linux 的環境裡

3)將 LAN stress for iperf and dp_tcp.zip 解壓縮

4)打開 LAN_stress_linux 資料夾

5)LAN_stress_linux.sh

6)進行 iperf 測試

Server 端

A.根據圖片十一可得知選擇"1"即可進行 iperf 測試

B.根據圖片十一可得知選擇"1"為 Server 端

C.看到圖片十一畫面即為完成 Server 端設定

```
Activities Terminal Feb 21 17:55
root@rhel-server:~/Documents

File Edit View Search Terminal Help
If iperf test item, please enter 1
If DP_TCP test item, please enter 2
1
=====
Please follow the Server and Client ip settings
Server ip :192.168.x.1
Client ip:192.168.x.2
And please execute the Server first
=====
The system is Server(192.168.1.1), please enter 1
If Client(192.168.1.2), please enter 2
Please enter your select (1 or 2):
1
Starting iperf server...
Starting iperf server on 192.168.1.1 port 5001!
Starting iperf server on 192.168.2.1 port 5002!
Starting iperf server on 192.168.3.1 port 5003!
Starting iperf server on 192.168.4.1 port 5004!
Starting iperf server on 192.168.5.1 port 5005!
iperf servers started on 7 interfaces.
Close Server Process : (y/n) ? -----
Server listening on TCP port 5005
Server listening on TCP port 5004
TCP window size: 85.3 KByte (default)TCP window size: 85.3 KByte (default)
-----
Server listening on TCP port 5002
TCP window size: 85.3 KByte (default)
-----
Server listening on TCP port 5001
TCP window size: 85.3 KByte (default)
-----
Server listening on TCP port 5003
```

Client 端

A.根據圖片十二可得知選擇"1"即可進行 iperf 測試

B.根據圖片十二可得知選擇"2"為 Client 端


```
Activities Terminal Feb 21 17:56
root@rhel-server:~/Documents

File Edit View Search Terminal Help
LAN_stress linux.sh _rom
[root@rhel-server Documents]# sh LAN_stress_linux.sh
iperf and bwm-ng are installed
Please enter the test item you want
If iperf test item, please enter 1
If DP_TCP test item, please enter 2
1
=====
Please follow the Server and Client ip settings
Server ip :192.168.x.1
Client ip:192.168.x.2
And please execute the Server first
=====
The system is Server(192.168.1.1), please enter 1
If Client(192.168.1.2), please enter 2
Please enter your select (1 or 2):
2
Starting iperf client...
Please enter your test time (sec)
600
Please enter your parallel value (default 8):
Starting iperf client for 192.168.1.1 on port 5001!
# watch fast: "/org/gnome/desktop/interface/" (establishing: 0, active: 0)
# watch fast: "/org/gnome/settings-daemon/peripherals/mouse/" (establishing: 0, active: 0)
# watch fast: "/org/gnome/desktop/sound/" (establishing: 0, active: 0)
# watch fast: "/org/gnome/desktop/privacy/" (establishing: 0, active: 0)
# watch fast: "/org/gnome/desktop/wm/preferences/" (establishing: 0, active: 0)
# watch fast: "/org/gnome/settings-daemon/plugins/xsettings/" (establishing: 0, active: 0)
# watch fast: "/org/gnome/desktop/ally/" (establishing: 0, active: 0)
# watch established: "/org/gnome/desktop/interface/" (establishing: 1)
# watch established: "/org/gnome/settings-daemon/peripherals/mouse/" (establishing: 1)
# watch established: "/org/gnome/desktop/sound/" (establishing: 1)
# watch established: "/org/gnome/desktop/privacy/" (establishing: 1)
# watch established: "/org/gnome/desktop/wm/preferences/" (establishing: 1)
# watch established: "/org/gnome/settings-daemon/plugins/xsettings/" (establishing: 1)
# watch established: "/org/gnome/desktop/ally/" (establishing: 1)
# watch fast: "/org/gnome/terminal/legacy/" (establishing: 0, active: 0)
# watch fast: "/org/gnome/terminal/legacy/" (establishing: 0, active: 0)
```

C.輸入需測試時間(以秒為單位)

D.看到圖片十三畫面即為完成 Client 端設定

```
Activities Terminal Feb 21 17:56
Terminal

File Edit View Search Terminal Help

[ 7] 0.00-10.00 sec 280 MBytes 234 Mbits/sec
[ 3] 0.00-10.00 sec 920 MBytes 772 Mbits/sec
[ 1] 0.00-10.00 sec 278 MBytes 233 Mbits/sec
[ 8] 0.00-10.00 sec 278 MBytes 233 Mbits/sec
[SUM] 0.00-10.00 sec 3.23 GBytes 2.78 Gbits/sec
[ 2] 10.00-20.00 sec 279 MBytes 234 Mbits/sec
[ 4] 10.00-20.00 sec 770 MBytes 646 Mbits/sec
[ 5] 10.00-20.00 sec 304 MBytes 255 Mbits/sec
[ 6] 10.00-20.00 sec 291 MBytes 244 Mbits/sec
[ 7] 10.00-20.00 sec 286 MBytes 240 Mbits/sec
[ 3] 10.00-20.00 sec 1013 MBytes 850 Mbits/sec
[ 1] 10.00-20.00 sec 354 MBytes 297 Mbits/sec
[ 8] 10.00-20.00 sec 280 MBytes 234 Mbits/sec
[SUM] 10.00-20.00 sec 3.49 GBytes 3.00 Gbits/sec
[ 8] 20.00-30.00 sec 274 MBytes 230 Mbits/sec
[ 2] 20.00-30.00 sec 271 MBytes 227 Mbits/sec
[ 4] 20.00-30.00 sec 724 MBytes 607 Mbits/sec
[ 5] 20.00-30.00 sec 301 MBytes 253 Mbits/sec
[ 6] 20.00-30.00 sec 282 MBytes 237 Mbits/sec
[ 7] 20.00-30.00 sec 282 MBytes 237 Mbits/sec
[ 3] 20.00-30.00 sec 1.05 GBytes 900 Mbits/sec
[ 1] 20.00-30.00 sec 386 MBytes 324 Mbits/sec
[SUM] 20.00-30.00 sec 3.51 GBytes 3.01 Gbits/sec
[ 8] 20.00-30.00 sec 1.39 GBytes 1.19 Gbits/sec
[SUM] 20.00-30.00 sec 6.27 GBytes 5.39 Gbits/sec
[ 1] 20.00-30.00 sec 902 MBytes 757 Mbits/sec
[SUM] 20.00-30.00 sec 6.61 GBytes 5.68 Gbits/sec
[ 1] 20.00-30.00 sec 508 MBytes 420 Mbits/sec
[SUM] 20.00-30.00 sec 6.61 GBytes 5.68 Gbits/sec

# watch established: "/org/gnome/desktop/wm/preferences/" (establishing: 1)
# watch established: "/org/gnome/settings-daemon/plugins/xsettings/" (establishing: 1)
# watch established: "/org/gnome/desktop/ally/" (establishing: 1)
# watch fast: "/org/gnome/terminal/legacy/" (establishing: 0, active: 0)
```

E.測試結束後 log 會放置在"iperf_log"裡

7)進行 DP_TCP 測試

Server 端

A.根據圖片十四可得知選擇"2"即可進行 DP_TCP 測試

B.根據圖片十四可得知選擇"1"為 Server 端

C.看到圖片十四畫面即為完成 Server 端設定

```
suse-lantest:~/Documents/LAN_stress_linux # sh LAN_stress_linux.sh
iperf and bwm-ng are installed
Please enter the test item you want
If iperf test item, please enter 1
If DP_TCP test item, please enter 2
2
=====
Please follow the Server and Client ip settings
Server ip :192.168.x.1
Client ip:192.168.x.2
And please execute the Server first
=====
The system is Server(192.168.1.1), please enter 1
If Client(192.168.1.2), please enter 2
Please enter your select (1 or 2):
1
Starting dp-tcp server...
dp_tcpcmpx64 Server C8004
Copyright (c) 1989-2005 Siemens Business Services, Germany
+-----+
| Attention! This is a version including debugging features |
+-----+
dp_tcpcmpx64: stdout and stderr redirected to /tmp/dp_tcpcmpx64.9000
dp_tcpcmpx64 Server C8004
Copyright (c) 1989-2005 Siemens Business Services, Germany
+-----+
| Attention! This is a version including debugging features |
+-----+
dp_tcpcmpx64: stdout and stderr redirected to /tmp/dp_tcpcmpx64.9001
dp_tcpcmpx64 Server C8004
```

Client 端

A.根據圖片十五可得知選擇"2"即可進行 DP_TCP 測試

B.根據圖片十五可得知選擇"2"為 Client 端

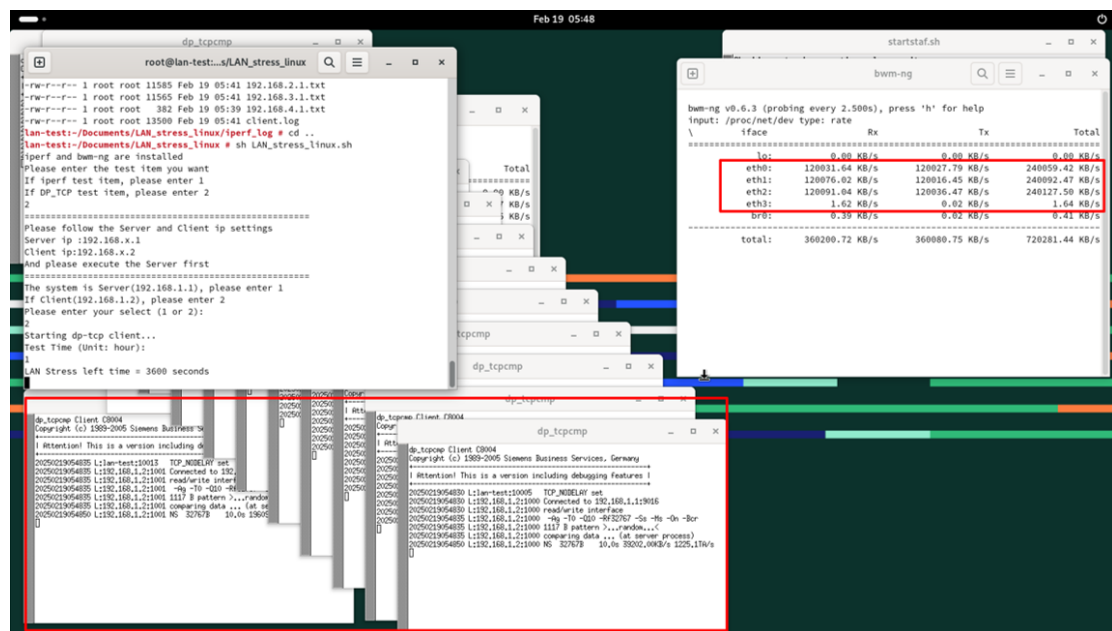
```

lan-test:~/Documents/LAN_stress_linux # sh LAN_stress_linux.sh
iperf and bwm-ng are installed
Please enter the test item you want
If iperf test item, please enter 1
If DP_TCP test item, please enter 2
2
=====
Please follow the Server and Client ip settings
Server ip :192.168.x.1
Client ip:192.168.x.2
And please execute the Server first
=====
The system is Server(192.168.1.1), please enter 1
If Client(192.168.1.2), please enter 2
Please enter your select (1 or 2):
2
Starting dp-tcp client...
Test Time (Unit: hour):
1
LAN Stress left time = 3600 seconds

```

C.輸入需測試時間(以小時為單位)

D.看到圖片十六畫面即為完成 Client 端設定



E.測試結束後 log 會放置在"dp_tcp_log"裡